

Leveraging Large Language Models for Anomaly Detection of Cyber Advanced Persistent Threats

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Abstract

Advanced Persistent Threats (APTs) are long-term, stealthy attacks that evade traditional signature-based detection. This project presents a dual-pipeline architecture that combines LLM semantic embeddings with Autoencoder-based anomaly detection and mathematically rigorous XAI attribution.

17 embedding models × 11 Autoencoder topologies benchmarked on DARPA TC adversarial datasets under strict anti-leakage protocol.

The framework achieves up to AUC-ROC 0.9997 while providing token-level SHAP attribution and a novel latent space ablation study that mathematically links human-readable threat tokens to the LLM's internal representations.

Problem Statement and Objectives

APTs operate by blending into normal system activity using living-off-the-land (LotL) techniques. Four specific problems motivate this work:

- No systematic embedding model comparison exists for APT log data across architectural families
- Autoencoder architecture diversity is limited; probabilistic and sequence-aware variants untested together
- SHAP is rarely applied to deep unsupervised detectors, only to supervised classifiers
- No explainable attribution bridges token-level text and 768-D LLM latent space representations

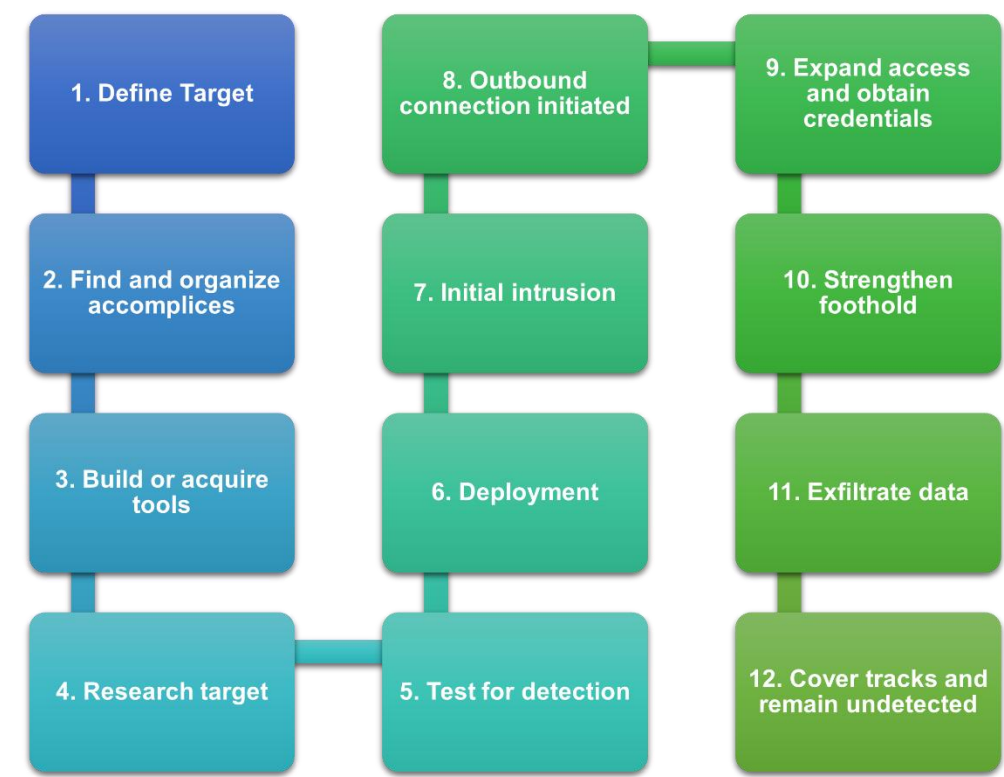


Figure 1.1: Stages of APT Attack

The Advanced Persistent Threat (APT) lifecycle begins when attackers select a target, gather intelligence, prepare tools, and test their methods to avoid detection. They then deploy the attack, gain initial access, establish outbound communication, and expand privileges while maintaining persistence inside the compromised environment. Finally, attackers exfiltrate sensitive data and hide their activities to remain undetected for as long as possible.

Literature Review

- APT-LLM (2026):** Used transformer embeddings such as BERT, ALBERT, and RoBERTa for APT anomaly detection, showing strong performance on DARPA TC datasets [1].
- LogShield (2023):** Applied RoBERTa with temporal log relationships, proving that sequence context improves cyber threat detection accuracy [2].
- OCR-APT (2025):** Combined Graph Neural Networks with LLM reasoning to support both attack detection and security investigation [3].
- Research Gap:** Most previous studies focus on detection accuracy, with limited work on unsupervised explainability and latent-space interpretation.

Research Methodology

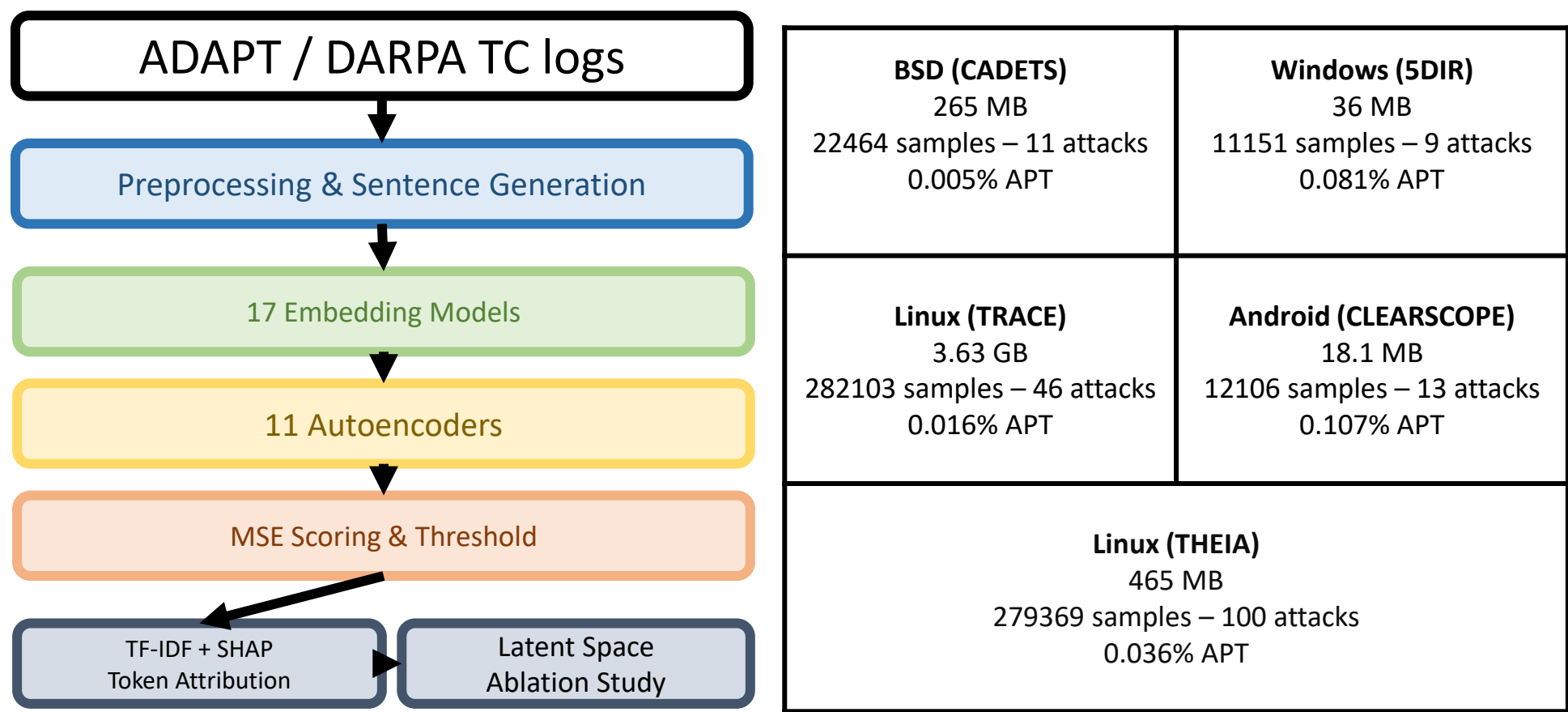


Figure 2: Methodology Steps

Table 1: DARPA TC E3 Dataset Overview

Dataset Overview & Experimental Setup

ADAPT repository (DARPA Transparent Computing). Five performers across four operating systems.

Anti-leakage split: 80% benign-only training. Zero APT events seen during training.

16 LLM models across 5 embedding groups: Encoder (BERT-based), Decoder (LLM-based), Large, Angle-Optimized, and Static. Hardware: NVIDIA RTX 1080 Ti + NVIDIA RTX 5060 Ti (CUDA).

Results and Discussions

0.9997

Peak AUC-ROC
(**contriever** + Baseline AE)

0.9421

Avg AUC-ROC
Vanilla AE

0.649

Best decoder
AUC-ROC

Encoder family (Group A) vastly outperforms decoder-based models. Instruction-tuned decoders near random chance on structured telemetry.

SHAP token attribution – Anomaly sample for 5DIR:

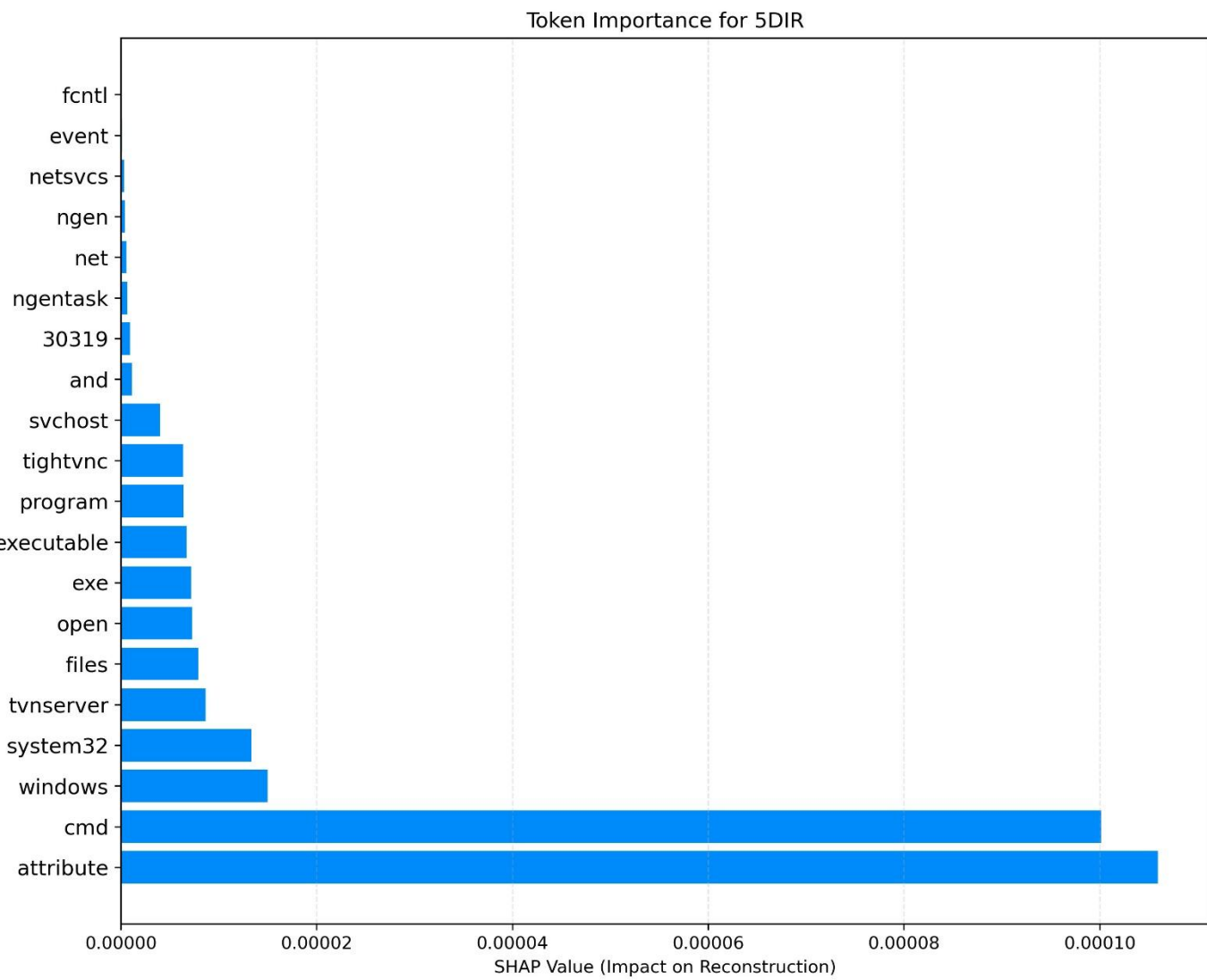


Figure 3: SHAP results for an anomaly instance

Ablation Study Results

Novel multi-stage ablation: SHAP-identified tokens are progressively masked and the log is re-embedded. MSE drop confirms each token's causal role in the LLM latent space.

PERFORMER	TYPE	ANOMALY	MEAN SHAP	RECONSTRUCTI ON ERROR
5DIR	Original	Process has event check_file_attributes and event close and ... and executable cmd.exe.1 and attribute driverquery	2.25e-07	0.0004
	Removed words	cmd, attribute	3.593e-08	0.0002

Anomalous logs confirm the analogy: masking **cmd**, **attribute** causes MSE to reduce from 0.0004 → 0.0002, proving the model anchors its anomalous classification to specific structural tokens.

Conclusion

- Encoder-based dense retrieval models far outperform instruction-tuned decoders on structured system telemetry. AUC-ROC gap: 0.907 vs 0.649.
- TF-IDF + SHAP pipeline successfully identifies randomized GUID fragments and transient execution paths as primary APT indicators, without any prior attack signatures.
- Latent space ablation mathematically proves the LLM's 768-D representations shift toward the benign distribution when threat tokens are removed. Black box pierced.

Future work: Graph Neural Network integration for process trees, real-time Kafka streaming deployment, and Agentic LLM remediation using SHAP outputs to auto-map detections to MITRE ATT&CK tactics.

References

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